

FVCOM-MP: Mass Conservation Bug Diagnosis

Delaware River Estuary Test Cases

WaterPACT FVCOM-MP Development

May 2026

Delaware River Test Cases

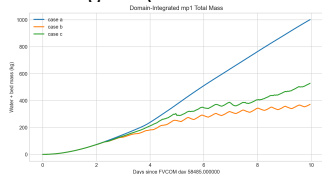
	Case a	Case b	Case c
PLAST_FLOC	F	T	T
SEDIMENT_MODEL	F [†]	T	T
SED_SD50	—	0.1 mm	1.0 mm
MP fields	conc	conc_a+conc_d	
Agg. gate d_{MP}^{max}	—	34 μm (excl.)	162 μm (act.)

[†]sediment initialized; adv_scal/FCT bypassed

- $\text{SOURCE_AG_RATE} = 0 \Rightarrow \text{conc_a} \equiv 0$
- $\rho_p = 950 \text{ kg m}^{-3}$,
 $d_p = 100 \mu\text{m}$

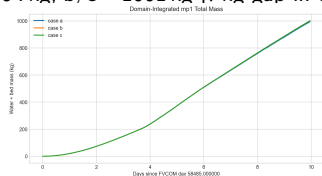
Mass Conservation Progress: Two Bugs, Two Fixes

v001 — original (full sediment active)



v007 — Fix B+ on b/c only

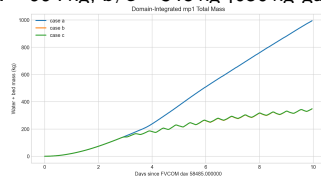
a = 994 kg; b/c = 1001 kg [7 kg gap in case a]



Bug 1: sed adv_scal/FCT corrupts shared scalar arrays
⇒ bypass sed horizontal adv/FCT

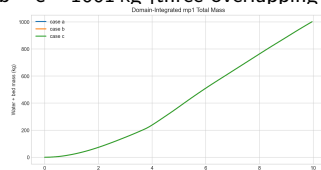
v006 — sed adv/FCT bypassed; NaN bug isolated

a = 994 kg; b/c = 348 kg [650 kg gap]



v008 — Fix B+ on all three ✓ SOLVED

a = b = c = 1001 kg [three overlapping lines]



Bug 2: rho_water = NaN → eflx = NaN → neg-clamp
destroys mass ⇒ isnan sentinel on eflx/pflx

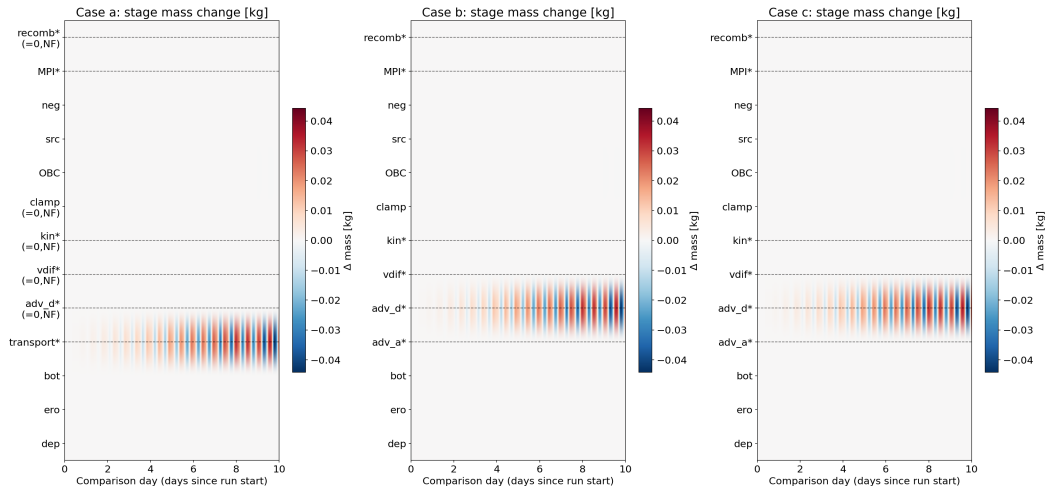
In-Run Mass Checkpoint Framework

CP	Stage	ΔM expected
0	start of timestep	<i>reference</i>
1–3	dep / ero / bed update	$\Delta(\text{water} + \text{bed}) = 0$
3p5	adv(conc_a) only	0* (conc_a = 0 now)
4	adv(conc_d)	OBC tidal exchange
5	vdif	0*
6	kinetics	0* (conc_a = 0 now)
7	upper clamp	0
8	OBC	0 (absorbed in adv)
9	PTsource (river)	river import
10	neg-clamp	0 (loss only)
11	MPI halo sync	0*
12	end of timestep	verify $\approx$ CP0

* dashed row in heatmap — must be exactly zero

Stage-Delta Heatmap: Fix B+ Active, All Cases, 10-day Run

Stage mass-change heatmap (diverging: red = gain, blue = loss)
Dashed lines = stages that must be exactly conservative



Grand-Total Mass Timeseries: CP0 vs. CP12 per Step

